

Multiple Choice

1. **Answer:** C

Options: A) Vitamin A; B) Vitamin D; C) Vitamin E; D) Vitamin K

Note: Vitamin E is a major lipid-soluble antioxidant in cell membranes because it protects cells from oxidative damage by neutralizing free radicals, thus maintaining cellular integrity and function.

2. **Answer:** C

Options: A) BMI > 25 kg/m²; B) BMI = 25 kg/m²; C) BMI 25-29.9 kg/m²; D) BMI 25-30 kg/m²

Note: The correct answer is based on the World Health Organization's classification of body mass index (BMI), which defines overweight as a BMI range of 25 to 29.9 kg/m², indicating a higher than normal weight relative to height that may pose health risks.

3. **Answer:** A

Options: A) All babies should be exclusively breastfed until around 6 months; B) Solids should be introduced at 4 months of age; C) Cow's milk is appropriate for all children from 6 months; D) Bottle feeding is more appropriate for babies than breast feeding

Note: Current recommendations emphasize that exclusive breastfeeding for the first six months provides optimal nutrition and supports healthy growth and development in infants.

4. **Answer:** A

Options: A) Fluoxetine is considered of benefit; B) The first approach is to discuss body image; C) It is not useful to discuss weight with patients; D) Treatments should target exclusively on purging practices

Note: Fluoxetine, a selective serotonin reuptake inhibitor, has been shown in clinical studies to effectively reduce binge-eating and purging behaviors in individuals with Bulimia Nervosa, making it a beneficial treatment option.

5. **Answer:** A

Options: A) First trimester of pregnancy.; B) Second trimester of pregnancy.; C) Third trimester of pregnancy.; D) Effects were similar at all stages of pregnancy.

Note: The first trimester of pregnancy is critical for fetal development, and research shows that exposure to nutritional stress during this period significantly alters DNA methylation patterns, which can have lasting impacts on health in the offspring.

True / False

6. **Answer:** True

Options: A) True; B) False

Note: A prospective study is considered the most reliable observational study for exploring the relationship between diet and cancer because it involves following participants over time, allowing researchers to observe changes in diet and subsequent cancer development in a temporal sequence, which helps establish cause-and-effect relationships more effectively than other study designs.

7. **Answer:** True

Options: A) True; B) False

Note: The food-price spike of 2007-2008 can be attributed to a combination of factors including the long-term underinvestment in agricultural infrastructure, the diversion of crops for biofuel production, and increased commodity speculation, which collectively disrupted supply and demand dynamics in the global food market.

8. **Answer:** False

Options: A) True; B) False

Note: Infants with severe combined immunodeficiency lack functional immune responses, making them highly vulnerable to infections and ineffective at responding to vaccinations.

9. **Answer:** False

Options: A) True; B) False

Note: Recent studies indicate that kwashiorkor is primarily associated with protein deficiency rather than low intake of antioxidant nutrients, making the statement false.

10. **Answer:** False

Options: A) True; B) False

Note: The statement is false because the WHO recommends limiting free sugars to less than 10% of total daily energy intake, not more than 10%.

Long Form Response

11. **Answer:** Genetically modified (GM) foods can significantly contribute to global food security and help address climate change by creating a more abundant and economical food supply. By using biotechnology, crops can be engineered to resist pests, tolerate extreme weather, and grow in less-than-ideal soil conditions, which helps increase yields and reduce losses. This not only ensures that more food is available to feed the growing population but also makes farming more sustainable in the face of climate change. Additionally, GM foods can be enhanced to improve their nutritional quality, providing essential vitamins and minerals that can help combat malnutrition in vulnerable communities. Overall, GM foods offer a promising solution to enhance food supply and nutritional standards while supporting efforts to mitigate climate challenges.

Note: correct because it highlights how GM foods enhance agricultural productivity

and resilience against climate challenges, ultimately increasing food availability and sustainability, which are crucial for addressing global food security.

12. **Answer:** Over the past two decades, China has experienced a rapid nutrition transition due to several factors, including urbanization, economic growth, and increasing globalization. As cities expand and income levels rise, people have shifted from traditional diets high in grains and vegetables to more Westernized diets rich in fats, sugars, and processed foods. This change has led to significant public health issues, such as rising obesity rates and chronic diseases like diabetes and heart disease. Additionally, the nutrition transition has influenced social dynamics, as dietary habits and food choices become symbols of modernity and status. Overall, while the transition reflects economic progress, it poses challenges for public health and requires strategies to promote healthier eating habits among the population.

Note: correct because it accurately identifies urbanization, economic growth, and globalization as key drivers of China's nutrition transition, while also highlighting the resulting public health challenges associated with dietary changes.